

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE-MEF-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E6

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

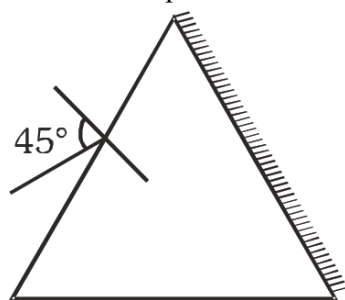
Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

Your Target is to secure Good Rank in Pre-Medical 2024

SECTION - A (PHYSICS)

1. One face of a prism with a refracting angle 30° is coated with silver. A ray is incident on another face at an angle of 45° is refracted and reflected from the silver coated face & retraces its path. The refractive index of prism is



- (1) $\sqrt{2}$ (2) 2
(3) $\sqrt{3}$ (4) 1.5
2. A concave mirror gives an image three times as large as the object placed at a distance of 20 cm from it. For the image to be virtual, the focal length should be -
- (1) -10 cm (2) -15 cm
(3) -20 cm (4) -30 cm
3. To a fish in water, a bird in air appears to be at 30 cm from the surface. If ${}_a\mu_w = \frac{4}{3}$, then the true distance of the bird from the surface is :-
- (1) 40 cm
(2) 22.5 cm
(3) 80 cm
(4) None of the above
4. A convex lens of focal length 40 cm is kept in contact with a concave lens of focal length 25 cm. The power of the combination is :
- (1) -6.5 D (2) +6.5 D
(3) -1.5D (4) +1.5D

5. Pick the **wrong** answer in the context with rainbow.

- (1) When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed.
(2) The order of colours is reversed in the secondary rainbow.
(3) An observer can see a rainbow when his front is towards the sun.
(4) Rainbow is a combined effect of dispersion refraction and reflection sunlight.

6. When a concave mirror of focal length 'f' is immersed in water, its focal length becomes 'f₀' then :-

- (1) $f_0 = f$ (2) $f_0 < f$
(3) $f_0 > f$ (4) Data insufficient

7. A thin equiconvex lens of power P is cut into 3 parts A, B & C as shown. If P₁, P₂ & P₃ are powers of the three parts respectively, then :-



- (1) $P_1 = P_2 = P_3$ (2) $P_1 > P_2 = P_3$
(3) $P_1 < P_2 = P_3$ (4) $P_2 = P_3 = 2P_1$

8. If focal length of objective lens is increased, then magnifying power of :-

- (1) Microscope will increase but that of telescope decrease.
(2) Microscope & telescope, both will increase.
(3) Microscope & telescope, both will decrease.
(4) Microscope will decrease but that of telescope will increase.

9. The angle of minimum deviation for a glass prism of refractive index $\sqrt{3}$ equals the refracting angle of the prism. The angle of prism is :-

(1) 30° (2) 60°
(3) 90° (4) 45°

10. A boy with defective eye-sight cannot see things beyond 50 cm. The corrective lens required has the power :

(1) +1D (2) +2D
(3) -1D (4) -2D

11. A ball is thrown upward and reaches a height of 64 feet, its initial velocity should be ($g = 32 \text{ ft/sec}^2$)

(1) 64 ft/sec (2) 72 ft/sec
(3) 32 ft/sec (4) 4096 ft/sec

12. Displacement of a particle as a function of time 't' is given by $s = \alpha + \beta t + \gamma t^2 + \delta t^4$ where α , β , γ and δ are constants. Ratio of initial velocity to initial acceleration is :-

(1) $\frac{\alpha}{\beta}$ (2) $\frac{\beta}{\gamma}$ (3) $\frac{2\beta}{\gamma}$ (4) $\frac{\beta}{2\gamma}$

13. Mark the correct statement.

- (1) If speed of a body is varying, its velocity must be varying and it must have zero acceleration
(2) If velocity of a body is varying, its speed must be varying
(3) A body moving with varying velocity may have constant speed
(4) A body moving with varying speed may have constant velocity if its direction of motion remains constant

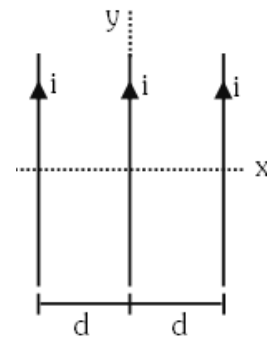
14. A train has a speed of 60 km/h for the first one hour and 40 km/h for the next half hour. Its average speed in km/h is :

(1) 50 (2) 53.33 (3) 48 (4) 70

15. A cylindrical conductor of radius 'R' carries a current 'i'. The value of magnetic field at a point which is R/4 distance inside from the surface is 10 T. Find the value of magnetic field at point which is 4R distance outside from the surface

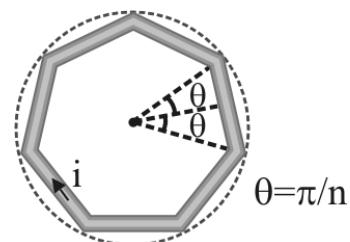
(1) $\frac{4}{3}T$ (2) $\frac{8}{3}T$ (3) $\frac{40}{3}T$ (4) $\frac{80}{3}T$

16. Three long parallel wires carrying same current i in the x-y plane, the middle wire is along y-axis as shown in the figure. Now the middle wire is displaced by distance z along z-axis, so the magnetic force per unit length acting on it is



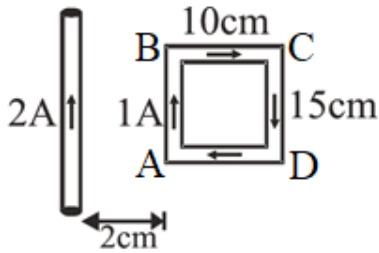
(1) $\frac{\mu_0 i^2 z}{2\pi(d^2 + z^2)}$ (2) $\frac{\mu_0 i^2 z}{\pi(d^2 + z^2)}$
(3) $\frac{\mu_0 i^2}{2\pi(d^2 + z^2)^{1/2}}$ (4) $\frac{\mu_0 i^2}{\pi(d^2 + z^2)^{1/2}}$

17. In the following figure a wire bent in the form of a regular polygon of n sides is inscribed in a circle of radius a. Net magnetic field at centre will be :-



(1) $\frac{\mu_0 i}{2\pi a} \tan \frac{\pi}{n}$
(2) $\frac{\mu_0 n i}{2\pi a} \tan \frac{\pi}{n}$
(3) $\frac{n\mu_0 i}{\pi a} \sin\left(\frac{\pi}{n}\right) \tan\left(\frac{\pi}{n}\right)$
(4) $\frac{n i}{2a} \mu_0 \tan \frac{\pi}{n}$

18. What is the net force on the square coil ?



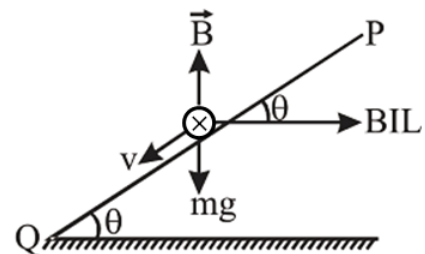
- (1) 25×10^{-7} N moving towards wire
 (2) 25×10^{-7} N moving away from wire
 (3) 35×10^{-7} N moving towards wire
 (4) 35×10^{-7} N moving away from wire
19. The acceleration of an electron at a moment in a magnetic field $\vec{B} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ is $\vec{a} = x\hat{i} - 2\hat{j} + \hat{k}$. The value of x is :-
- (1) 0.5 (2) 1
 (3) 2.5 (4) 1.5
20. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If E and B represent the electric and magnetic fields respectively, then this region of space may have :-
- (i) $E = 0, B = 0$ (ii) $E = 0, B \neq 0$
 (iii) $E \neq 0, B = 0$ (iv) $E \neq 0, B \neq 0$
- Correct statements are –
- (1) i, ii and iii (2) ii, iii and iv
 (3) i, ii and iv (4) All
21. A helium nucleus makes full rotation in a circle of radius 0.8 m in 2 s. The value of magnetic field B at the centre of the circle, will be (μ_0 = permeability constant) :-

- (1) $\frac{2 \times 10^{-19}}{\mu_0}$
 (2) $2 \times 10^{-19} \mu_0$
 (3) $10^{-19} \mu_0$
 (4) $\frac{10^{-19}}{\mu_0}$

22. When a proton is released from rest in a room, it starts with an initial acceleration a_0 towards north. When it is projected towards east with a speed v_0 it moves with an initial acceleration $3a_0$ towards south. The electric and magnetic fields in the room are :

- (1) $\frac{ma_0}{e}$ east, $\frac{2ma_0}{ev_0}$ up
 (2) $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ down
 (3) $\frac{ma_0}{e}$ north, $\frac{4ma_0}{ev_0}$ up
 (4) $\frac{ma_0}{e}$ south, $\frac{4ma_0}{ev_0}$ down

23. A conducting rod of length L and mass m is moving down a smooth inclined plane of inclination θ with constant speed v . A current I is flowing in the conductor perpendicular to the paper inwards. A vertically upward magnetic field B exists in space there. The magnitude of magnetic field $\vec{B}$ is :-



- (1) $\frac{mg}{IL} \sin \theta$ (2) $\frac{mg}{IL} \cos \theta$
 (3) $\frac{mg}{IL} \tan \theta$ (4) $\frac{mg}{IL} \sec \theta$
24. A dip needle lies initially in the magnetic meridian when it shows an angle of dip θ at a place. The dip circle is rotated through an angle x in the horizontal plane and then it shows an angle of dip θ' . Then $\frac{\tan \theta'}{\tan \theta}$ is :
- (1) $\frac{1}{\cos x}$ (2) $\frac{1}{\sin x}$
 (3) $\frac{1}{\tan x}$ (4) $\cos x$

25. A bar magnet is oscillating in the Earth's magnetic field with a period T . What happens to its period and motion if its mass is quadrupled? [Keeping pole strength and dimension of magnet same].

- (1) Motion remains S.H.M. with time period $= \frac{T}{2}$.
 (2) Motion remains S.H.M. with time period $= 2T$.
 (3) Motion remains S.H.M. with time period $= 4T$.
 (4) Motion remains S.H.M. and period remains nearly constant.

26. To increase the current sensitivity of a moving coil galvanometer, we should :-

- (1) decrease number of turns in coil
 (2) decrease area of cross section of the coil
 (3) increase torsional constant of spiral springs
 (4) None of the above

27. The root mean square velocity of the molecules in a sample of helium is $\frac{1}{\sqrt{2}}$ times of that of the molecules in a sample of hydrogen. If the temperature of the hydrogen sample is 0°C , that of helium samples is approximately is about :-

- (1) 0°C (2) 0 K
 (3) 273°C (4) 100°C

28. An ideal gas having molar specific heat capacity at constant volume is $\frac{3}{2}R$, the molar specific heat capacity at constant pressure is:-

- (1) $\frac{1}{2}R$ (2) $\frac{5}{2}R$ (3) $\frac{7}{2}R$ (4) $\frac{9}{2}R$

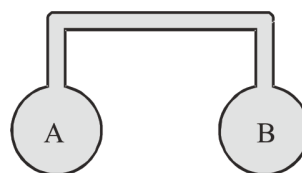
29. If one mole of a monoatomic gas $\left(\gamma = \frac{5}{3}\right)$ is mixed with one mole of a diatomic gas $\left(\gamma = \frac{7}{5}\right)$, the value of γ for the mixture is :-

- (1) 1.4 (2) 1.5 (3) 1.53 (4) 3.07

30. The internal energy of 2 moles of a monoatomic gas at temp. T is :-

- (1) $\frac{3}{2}RT$ (2) $3RT$
 (3) $2RT$ (4) $5RT$

31. Two spherical vessel of equal volume, are connected by a narrow tube. The apparatus contain an ideal gas at one atmosphere and 300 K . Now if one vessel is immersed in a bucket of constant temperature 600 K and the other in a bucket of constant temperature 300 K . Then the common pressure will be :



- (1) 1 atm (2) $\frac{4}{5}\text{ atm}$
 (3) $\frac{4}{3}\text{ atm}$ (4) $\frac{3}{4}\text{ atm}$

32. For the molecules of an Ideal gas, Which of the following velocity average can not be zero

- (1) $\langle v \rangle$ (2) $\langle v^4 \rangle$ (3) $\langle v^3 \rangle$ (4) $\langle v^5 \rangle$

33. One mole of an ideal gas at temperature T_1 expands according to the law $\frac{P}{V^2} = a$ (constant). The work done by the gas till temperature of gas becomes T_2 is :

- (1) $\frac{1}{2}R(T_2 - T_1)$ (2) $\frac{1}{3}R(T_2 - T_1)$
 (3) $\frac{1}{4}R(T_2 - T_1)$ (4) $\frac{1}{5}R(T_2 - T_1)$

34. The molecules of a given mass of a gas have r.m.s. velocity of 200 m/s at 27°C and $1.0 \times 10^5\text{ N/m}^2$ pressure. When the temperature and pressure of the gas are respectively, 127°C and $0.05 \times 10^5\text{ N/m}^2$, the r.m.s. velocity of its molecules in m/s is :

- (1) $100\sqrt{2}$ (2) $\frac{400}{\sqrt{3}}$
 (3) $\frac{100\sqrt{2}}{3}$ (4) $\frac{100}{3}$

35. Relation $PV = nRT$ is given for following condition of real gas -

- (1) High temperature and high density
- (2) Low temperature and low density
- (3) High temperature and low density
- (4) Low temperature and high density

SECTION - B (PHYSICS)

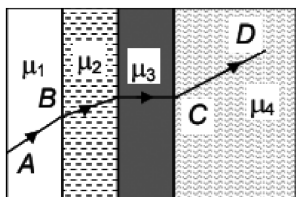
36. A ray of light is incident at angle i on a surface of a prism of small angle A and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , the angle of incidence i is nearly equal to :

- (1) A/μ (2) $A/(2\mu)$
- (3) μA (4) $\mu A/2$

37. A concave mirror and a converging lens (glass with $\mu = 1.5$) both have a focal length of 3 cm when in air. When they are in water ($\mu = \frac{4}{3}$), their new focal lengths are

- (1) $f_{\text{Lens}} = 12 \text{ cm}$, $f_{\text{Mirror}} = 3 \text{ cm}$
- (2) $f_{\text{Lens}} = 3 \text{ cm}$, $f_{\text{Mirror}} = 12 \text{ cm}$
- (3) $f_{\text{Lens}} = 3 \text{ cm}$, $f_{\text{Mirror}} = 3 \text{ cm}$
- (4) $f_{\text{Lens}} = 12 \text{ cm}$, $f_{\text{Mirror}} = 12 \text{ cm}$

38. A ray of light passes through four transparent media with refractive indices μ_1, μ_2, μ_3 and μ_4 as shown in the figure. The surface of all media are parallel. If the emergent ray CD is parallel to the incident ray AB, we have:

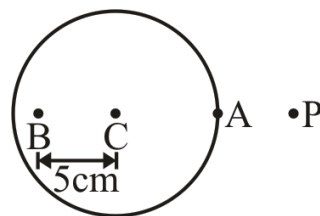


- (1) $\mu_1 = \mu_2$ (2) $\mu_2 = \mu_3$
- (3) $\mu_3 = \mu_4$ (4) $\mu_4 = \mu_1$

39. Speed of light in media M_1 & M_2 is $1.5 \times 10^8 \text{ m/s}$ and $2 \times 10^8 \text{ m/s}$ respectively. A ray of light enters from medium M_1 to M_2 at an angle of incidence ' i '. If the ray suffers total internal reflection, the value of ' i ' is :-

- (1) $\sin^{-1}\left(\frac{2}{3}\right)$ (2) $\leq \sin^{-1}\left(\frac{3}{5}\right)$
- (3) $> \sin^{-1}\left(\frac{3}{4}\right)$ (4) $< \sin^{-1}\left(\frac{2}{3}\right)$

40. In a glass ($\mu = 1.5$) sphere with a radius of 10 cm, there is an air bubble B at a distance of 5 cm from C. The distance of the bubble from point A, as observed from point P in air, will be :-



- (1) 4.5 cm (2) 20 cm
- (3) 9.37 cm (4) 6.67 cm

41. A particle starts from rest, accelerates at 2 m/s^2 for 10 s and then goes with constant speed for 30 s and then decelerates at 4 m/s^2 till it stops. What is the distance travelled by it.

- (1) 750 m (2) 1900 m
- (3) 100 m (4) 1800 m

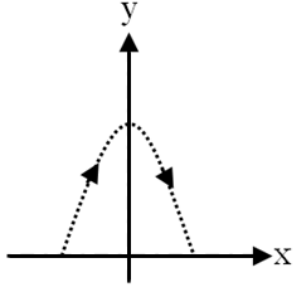
42. There are four light-weight-rod samples A, B, C and D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted :-

- (i) A is feebly repelled
- (ii) B is feebly attracted
- (iii) C is strongly attracted
- (iv) D remains unaffected

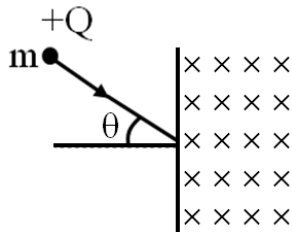
Which one of the following is true ?

- (1) C is of a diamagnetic material
- (2) D is of a ferromagnetic material
- (3) A is of a non-magnetic material
- (4) B is of a paramagnetic material

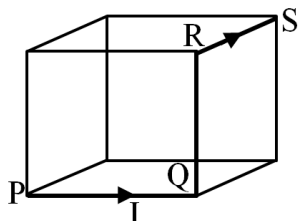
43. A wire carrying a current of 4A is bent in the form of a parabola $x^2 + y = 16$ as shown in figure, where x and y are in meter. The wire is placed in a uniform magnetic field $\vec{B} = 5\hat{k}$ tesla. The force acting on the wire is :



- (1) $80\hat{j}$ N (2) $-80\hat{j}$ N
(3) $-160\hat{j}$ N (4) $160\hat{j}$ N
44. A particle with charge $+Q$ and mass m enters in a magnetic field of magnitude B as shown in figure. The time spent by the particle in the field will be (where $T = \frac{2\pi m}{QB}$)



- (1) $T\theta$ (2) $2T\theta$
(3) $T \left(\frac{\pi + 2\theta}{2\pi} \right)$ (4) $T \left(\frac{\pi - 2\theta}{2\pi} \right)$
45. A wire PQRS carrying a current I runs along three edges of a cube of side ℓ as shown. There exists a uniform magnetic field of magnitude B along one of the sides of the cube. The magnitude of the force acting on the wire is :



- (1) 0 (2) $\sqrt{3}I\ell B$
(3) $\sqrt{2}I\ell B$ (4) $2I\ell B$

46. A charged particle enters in a magnetic field at right angles to the magnetic field. The field exists for a length equal to 1.5 times the radius of circular path of the particle. The particle will be deviated from its path by :

- (1) 90° (2) $\sin^{-1} \left(\frac{2}{3} \right)$
(3) 30° (4) 180°

47. One mole of an ideal monoatomic gas at temperature T_0 expands slowly according to law $P = kV$ (k is constant). If final temperature is $2T_0$, heat supplied to gas is :-

- (1) $2RT_0$ (2) $\frac{3}{2}RT_0$
(3) RT_0 (4) $RT_0/2$

48. 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the same initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :

- (1) 32 kJ (2) 5 kJ
(3) -4 kJ (4) 13 kJ

49. A gas has molar heat capacity $C = 4.5R$ in the process $PT = \text{constant}$. The number of degrees of freedom of molecules in the gas is :-

- (1) 4 (2) 3
(3) 6 (4) 5

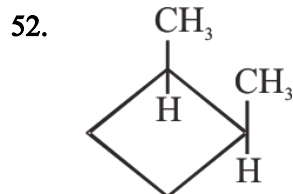
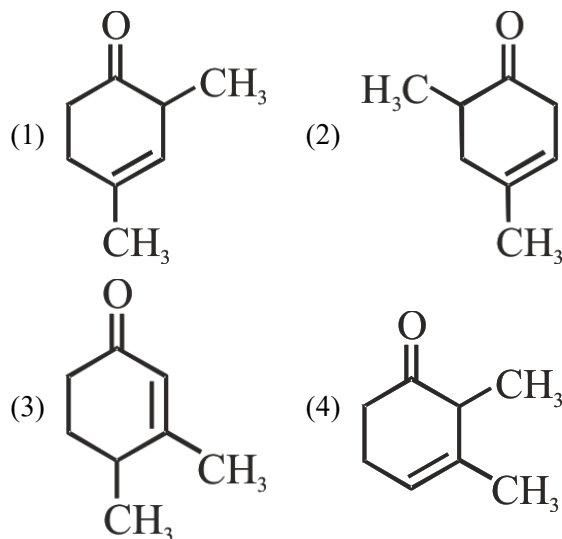
50. A gas is found to obey the law $PV^{2/3} = \text{constant}$. The initial temperature and volume are T_0 and V_0 . If gas expand to a volume $8V_0$ then final temperature becomes :-

- (1) T_0 (2) $\sqrt{2}T_0$
(3) $2T_0$ (4) $8T_0$

Topic : POLYMERS, OXYGEN CONTAINING COMPOUNDS, AMINES, BIOMOLECULE, CHEMISTRY IN EVERYDAY LIFE ENVIRONMENTAL CHEMISTRY, CHEMICAL BONDING (COMPLETE), CHEMICAL EQUILIBRIUM, IONIC EQUILIBRIUM (pH OF SA, SB MIXING) **[BACK UNIT] : NOMENCLATURE, ISOMERISM**

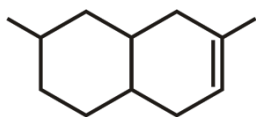
SECTION-A (CHEMISTRY)

51. 2,4-Dimethyl-3-cyclohexenone can be written as :-



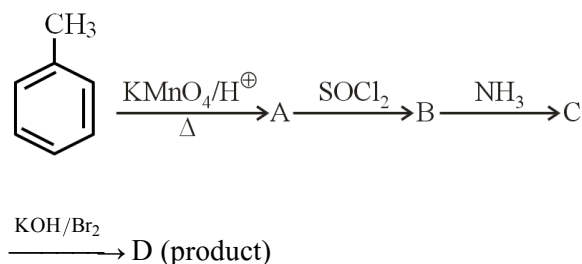
Incorrect statement about this compound is :-

- (1) It shows geometrical isomerism
 (2) It posses plane of symmetry
 (3) It posses center of symmetry
 (4) It shows optical isomerism
53. Dihedral angle in most stable conformer of butane is
- (1) 0° (2) 60°
 (3) 180° (4) None
54. How many chiral centre are present in the compound?

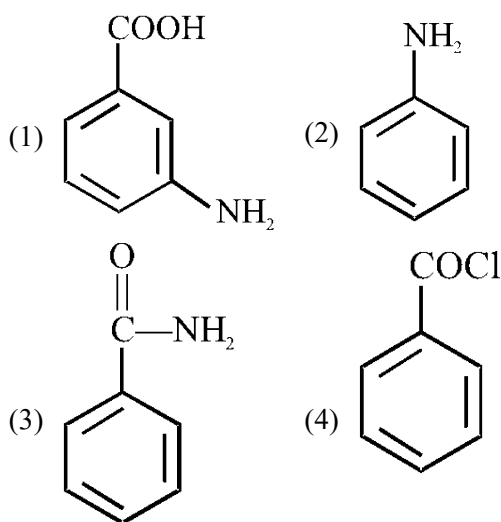


- (1) 1 (2) 2 (3) 3 (4) 4

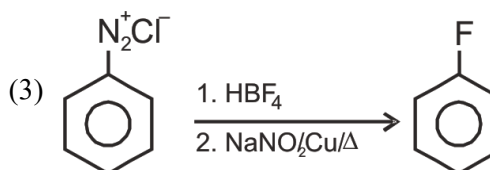
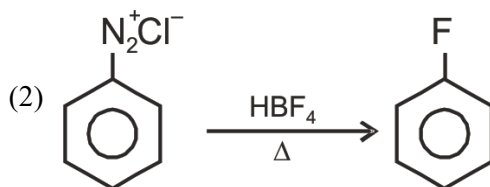
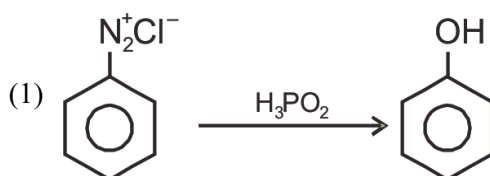
55. Suppose the following reaction -



The final product will be -

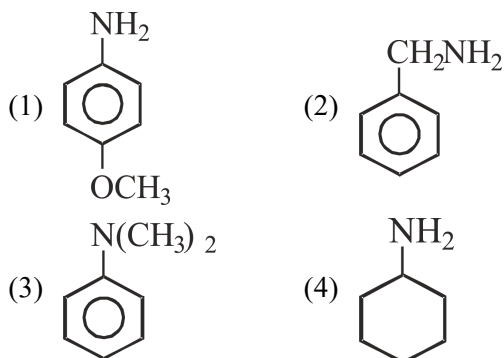


56. Identify the incorrect reaction(s).



- (4) Both (1) & (3)

57. Which of the following does not give carbylamine test on heating with CHCl_3 and Alc. KOH :-



58. In Kjeldahl's method, nitrogen is liberated as

- (1) $\text{N}_2(\text{g})$ (2) $\text{NO}(\text{g})$
(3) $\text{N}_2\text{O}(\text{g})$ (4) $\text{NH}_3(\text{g})$

59. Which of the following is an addition polymer?

- (1) Nylon 6 (2) Nylon 6, 6
(3) Buna S (4) Dacron

60. Which one of the following monomer gives the polymer neoprene on polymerization -

- (1) $\text{CH}_2 = \text{CHCl}$
(2) $\text{CCl}_2 = \text{CCl}_2$
(3) $\text{CH}_2 = \underset{\text{Cl}}{\text{C}} - \text{CH} = \text{CH}_2$
(4) $\text{CF}_2 = \text{CF}_2$

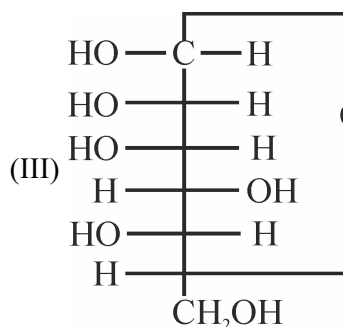
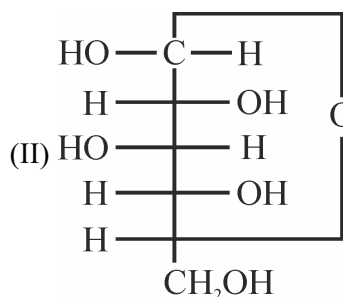
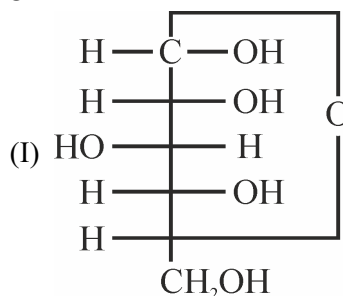
61. Which of the following statement is not true ?

- (1) The maximum limit of nitrate in drinking water is 50 ppm.
(2) Oxidation of sulphur dioxide into sulphur tri oxides is catalysed by particulate matter, ozone and hydrogen peroxide.
(3) Excess sulphates (> 500 ppm) in drinking water cause laxative effect.
(4) Common components of photochemical smog are ozone, nitric oxide and sulphur dioxide.

62. Which of the following statement is incorrect.

- (1) High density polythene is prepared in presence of Ziegler-Natta catalyst
(2) Terylene or dacron is the condensation polymer of ethylene glycol and terephthalic acid
(3) Penicillin G is narrow spectrum antibiotics while Ampicillin and Amoxicillin are broad spectrum antibiotics
(4) Iproniazid & phenelzine belong to the class of antihistamines drugs.

63. Three cyclic structures of monosaccharides are given below:



Which of these are anomers :

- (1) I and II (2) II and III
(3) I and III (4) I, II and III

64. Dettol is a mixture of :

- (1) Chloroxylonol and terpineol
- (2) Chloroxylonol and sulphur dioxide
- (3) Terpineol and sulphur dioxide
- (4) Phenol and terpienol

65. Consider the reaction $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ for which $K_c = 278 \text{ M}^{-1}$. If 0.001 mole of each of the reagents $\text{SO}_2(\text{g})$, $\text{O}_2(\text{g})$ and $\text{SO}_3(\text{g})$ are mixed in a 1.0 L flask. Determine the reaction quotient of the system and the spontaneous direction of the system:-

- (1) $Q_c = 1000$; the equilibrium shifts to the right
- (2) $Q_c = 1000$; the equilibrium shifts to the left
- (3) $Q_c = 0.001$; the equilibrium shifts to the left
- (4) $Q_c = 0.001$; the equilibrium shifts to the right

66. In the reaction $\text{X}(\text{g}) + \text{Y}(\text{g}) \rightleftharpoons 2 \text{Z}(\text{g})$, 2 mole of X, 1 mole of Y and 1 mole of Z are placed in a 10 litre vessel and allowed to reach equilibrium. If final concentration of Z is 0.2 M, then K_c for the given reaction is :-

- (1) 1.60
- (2) $\frac{80}{3}$
- (3) $\frac{16}{3}$
- (4) None of these

67. Given the following reaction at equilibrium, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$. At constant pressure if some inert gas is added to the system. Then :-

- (1) More $\text{NH}_3(\text{g})$ is produced
- (2) Less $\text{NH}_3(\text{g})$ is produced
- (3) No affect on the equilibrium
- (4) K_p of the reaction is decreased

68. At 444 °C, the equilibrium constant K for the reaction $2\text{AB}(\text{g}) \rightleftharpoons \text{A}_2(\text{g}) + \text{B}_2(\text{g})$ is $\frac{1}{64}$. The degree of dissociation of AB will be -

- (1) 10%
- (2) 20%
- (3) 30%
- (4) 50%

69. If $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$; $(p_{\text{CO}_2})_{\text{eq}} = 2 \text{ atm}$
 $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons 2\text{CO}(\text{g})$; $K_p = 6 \text{ atm}$
 Then K_p for the reaction :
 $\text{CaCO}_3(\text{s}) + \text{C}(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + 2\text{CO}(\text{g})$ is :

- (1) 12 atm
- (2) 6 atm
- (3) 12 atm^2
- (4) 6 atm^2

70. The unit of K_c for the reaction
 $\text{HCN}(\text{aq}) + \text{NaOH}(\text{aq}) \rightleftharpoons \text{NaCN}(\text{aq}) + \text{H}_2\text{O}(\ell)$

- (1) M
- (2) M^{-1}
- (3) M^{-2}
- (4) unit less

71. If 8 moles of $\text{PCl}_5(\text{g})$ is taken in vessel to react according to the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$, if at equilibrium total moles are 10 & equilibrium pressure is 5 atm then find k_p .

- (1) $\frac{2}{3} \text{ atm}$
- (2) $\frac{4}{3} \text{ atm}$
- (3) $\frac{1}{3} \text{ atm}$
- (4) $\frac{7}{3} \text{ atm}$

72. The ionic product of water at 50°C is 4×10^{-14} . The pH and nature of water is

- (1) 6.3, Acidic
- (2) 7, Neutral
- (3) 6.7, Neutral
- (4) 7.3, Basic

73. When 100 ml of 1 M NaOH solution is mixed with 10 ml of 10 M H_2SO_4 solution the resulting mixture will be

- (1) acidic
- (2) alkaline
- (3) neutral
- (4) strongly alkaline

74. Select the correct order of bond strength ?

- (1) $\text{C}-\text{C} < \text{Si}-\text{Si} < \text{Ge}-\text{Ge}$
- (2) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- (3) $\text{H}-\text{Cl} > \text{HBr} > \text{HI} > \text{HF}$
- (4) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$

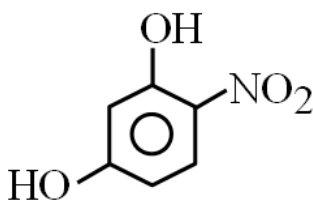
75. Which of following is non-polar molecule with presence of polar bonds and $\ell.P.$ at central atom.

- (1) CO_2 (2) SF_4 (3) XeF_4 (4) CF_4

76. Which is incorrect match ?

- (1) $\text{Na}^+ + \text{H}_2\text{O}$ dipole - dipole attraction
 (2) $\text{HCl} + \text{HCl}$ Keesome attraction
 (3) $\text{Xe} + \text{H}_2\text{O}$ Debye attraction
 (4) $\text{CCl}_4 + \text{CCl}_4$ London force

77. The type of molecular force of attraction present in the following is :-



- (1) Inter molecular H-bonding
 (2) Intra molecular H-bonding
 (3) dipole-dipole attraction
 (4) All of these

78. CO_2 is isostructural with -

- (a) BeCl_2 (b) SnCl_2
 (c) NO_2^- (d) C_2H_2

The correct answer is -

- (1) a, d (2) a, b
 (3) b, c (4) a, b, c

79. If Z-axis is the inter molecular axis, then π -molecular orbitals are formed by the overlap of -

- (1) $s + p_z$ (2) $p_x + p_y$
 (3) $p_z + p_z$ (4) $p_x + p_x$

80. The bond order of NO^+ is -

- (1) 3 (2) 2.5 (3) 2.0 (4) 1.5

81. Which of the following species is paramagnetic in nature ?

- (1) NO (2) CO (3) N_2 (4) O_2^{2-}

82. Shape of I_3^- & I_3^+ respectively :

- (1) T-shape and V-shape
 (2) Linear and T shape
 (3) Both have T shape
 (4) Linear & V shape

83. Hypervalent compound is :-

- (1) SO_3^{2-} (2) NH_3
 (3) BeF_2 (4) CH_4

84. A molecule which can not exist theoretically is :-

- (1) SF_4 (2) OF_2 (3) OF_4 (4) O_2F_2

85. Match List - I with List - II.

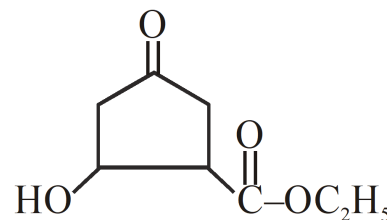
List-I	List-II
(a) PCl_5	(i) Square pyramidal
(b) SF_6	(ii) Trigonal planar
(c) BrF_5	(iii) Octahedral
(d) BF_3	(iv) Trigonal bipyramidal

Choose the correct answer from the options given below.

- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

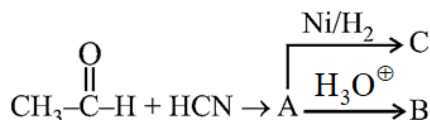
SECTION-B (CHEMISTRY)

86. The correct IUPAC name of given compound is-

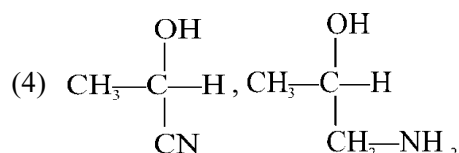
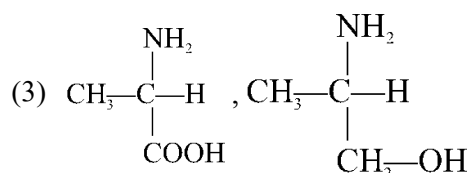
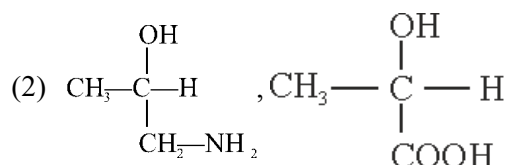
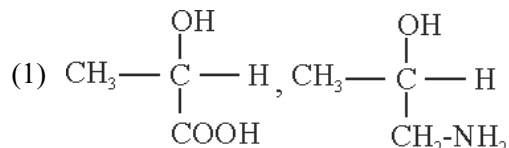


- (1) Ethyl-3-hydroxy-5-oxo cyclo hexanoate
 (2) Ethyl-2-hydroxy-4-oxo cyclo pentane carboxylate
 (3) Ethyl-5-hydroxy-3-oxo cyclo pentane carboxylate
 (4) Ethyl-6-hydroxy-4-oxo cyclo haxanoate

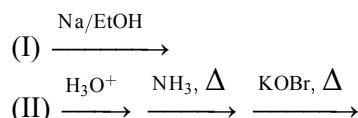
87. Suppose the following reaction -



The structure of B & C will be respectively :

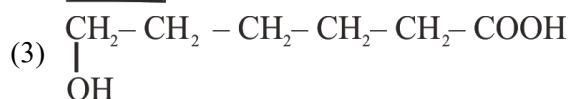
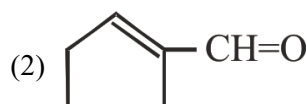
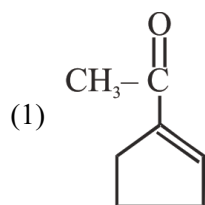


88. Ethylcyanide can be converted into ethylamine by

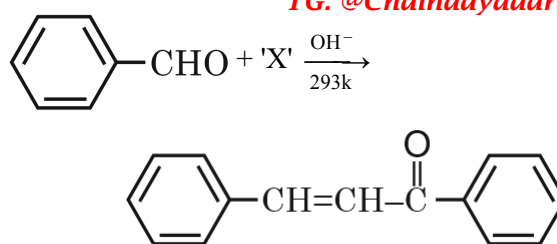


- (1) Only I (2) Only II
 (3) Both (I) and (II) (4) None of these

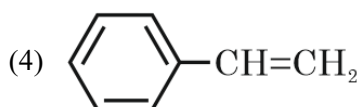
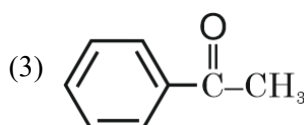
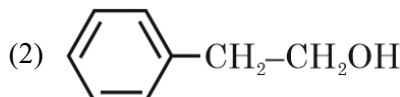
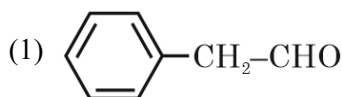
89. $\text{H}-\text{C}(=\text{O})-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})-\text{H} \xrightarrow{\text{OH}^-, \Delta} \text{A}$.
 Product A is



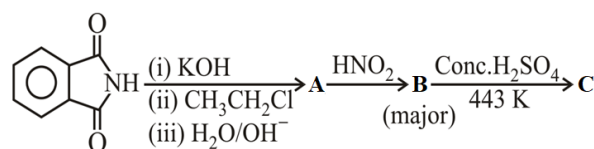
90.



Identify 'X'.



91.



Identify incorrect statement ?

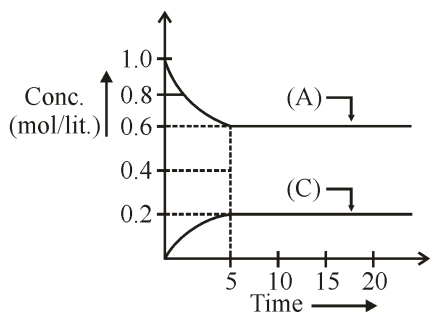
- (1) A gives mustard oil smell on reaction with CS_2 .
 (2) B gives red color in Victor Meyer test.
 (3) C is heated with Ziegler-Natta catalyst to form high density polyethylene.
 (4) When A reacts with Hinsberg reagent the product formed is insoluble in base.

92.

For an equilibrium, $\text{H}_2\text{O(s)} \rightleftharpoons \text{H}_2\text{O(l)}$ which of the following statement is true :-

- (1) The pressure changes do not affect the equilibrium
 (2) More of ice melts if pressure on the system is increased
 (3) More of liquid freezes if pressure on the system is increased
 (4) All of these

93. For the reaction $A \xrightleftharpoons[k_b]{k_f} C$, the concentration Vs time graph is shown as



what is the value of K_c for the reaction $2C \rightleftharpoons 2A$?

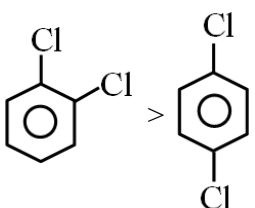
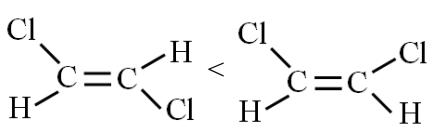
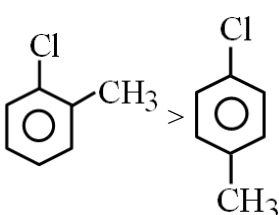
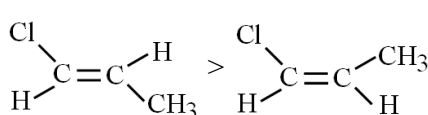
- (1) 4 (2) 9 (3) 6 (4) 7
94. $\alpha \propto \frac{1}{\sqrt{V}}$ is true for (where α is degree of dissociation) [consider $\alpha \ll 1$]

- (1) $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)}$
 (2) $2A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$
 (3) $A_{(g)} + 3B_{(g)} \rightleftharpoons C_{(g)}$
 (4) $A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$

95. The d-orbitals involved in sp^3d^2 hybridisation

- (1) $d_{x^2-y^2}, d_{xy}$ (2) $d_{z^2}, d_{x^2-y^2}$
 (3) d_{z^2}, d_{xz} (4) d_{xz}, d_{yz}

96. Which dipole moment order is incorrect ?

- (1) 
- (2) 
- (3) 
- (4) 

97. Which order of thermal stability is incorrect ?

- (1) $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4$
 (2) $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2$
 (3) $BeF_2 > MgF_2 > CaF_2 > SrF_2$
 (4) $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3$

98. Which is not correct condition of hybridisation ?

- (1) The orbitals present in the valence shell of the atom are hybridised
 (2) The orbitals undergoing hybridisation should have large energy difference
 (3) Promotion of electrons is not essential condition prior to hybridisation
 (4) Half filled/fully filled any type orbital can participate in hybridisation

99. Which is incorrect statement for molecular orbital theory.

- (1) The atomic orbitals of comparable energies and proper symmetry combine to form molecular orbitals.
 (2) Atomic orbital is polycentric and molecular orbital is monocentric.
 (3) The bonding molecular orbital has lower energy so higher stability.
 (4) Electron density in a bonding M.O. is located between the nuclei of the bonded atoms.

100. Which of following compound is formed in second excited state :-

- (1) PCl_5 (2) CH_4
 (3) SF_6 (4) IF_7

Topic : BIOTECH-APPLICATIONS, BIOMOLECULES, PLANT BREEDING, CELL BIOLOGY (COMPLETED), PLANT DIVERSITY, COMPLETE PLANT ANATOMY [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM & REPRODUCTION IN ORGANISMS.

SECTION-A (BIOLOGY-I)

101. GM plants have been useful in many ways as genetic modification has:

- (A) Made crops more tolerant to abiotic stresses
- (B) Reduced reliance on chemicals by making pesticide-resistant crops.
- (C) Helped to reduce post harvest losses.
- (D) Decreased efficiency of mineral usage by plants.
- (E) Enhanced nutritional value of food.

- (1) A,B,C,D & E (2) A,B,C & E
- (3) A,C,D & E (4) A,C & E

102. The proteins encoded by the gene cryIAb controls -

- (1) Corn borer
- (2) Cotton bollworm
- (3) Aphids
- (4) Jassids

103. PCR is not being used to detect which of the following -

- (1) HIV infection
- (2) Cancer related mutations
- (3) Cardiac diseases
- (4) Genetic disorders

104. A very time-consuming and tedious process in plant breeding programs is -

- (1) Collection of variability
- (2) Evaluation and selection of parents
- (3) Cross hybridization among selected parents
- (4) Selection and testing of superior recombinants

105. Assertion : lipids are monomeric substances found in acid insoluble pool.

Reason : lipids are in form of vesicles which are not water soluble.

- (1) Both **Assertion** and **Reason** are true and the **Reason** is correct explanation of **Assertion**.
- (2) Both **Assertion** and **Reason** are true but the **Reason** is not correct explanation of **Assertion**.
- (3) **Assertion** is true but **Reason** is false.
- (4) **Assertion** is false but **Reason** is true.

106. Which of the following is a Lectin?

- (1) Abrin (2) Concanavalin A
- (3) Carotenoids (4) Curcumin

107. RNAi takes place in _____ organisms as a method of cellular defense.

- (1) all eukaryotic (2) all prokaryotic
- (3) all non-living (4) some eukaryotic

108. Which organisation will make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services in India?

- (1) CDRI (2) GEAC
- (3) IRRI (4) ICAR

109. One long protein chain is folded upon itself like a hollow woolen ball, giving rise to the -

- (1) Primary structure
- (2) Secondary structure
- (3) Tertiary structure
- (4) Quaternary structure

110. Most abundant protein in whole biosphere is -

- (1) RuBisCo (2) Collagen
- (3) Cellulose (4) Chitin

TG: @Chalnaayaaar

PHASE - MEF

1015CMD303361230061

14-08-2023

111. Statement-1 :- Biopiracy is the term used to refer to the use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment.

Statement-2 :- In 1983, an American company got patent rights on Basmati rice through the US Patent and Trademark Office.

- (1) Both **Statement-1** and **Statement-2** are correct.
- (2) **Statement-1** is correct but **Statement-2** is incorrect.
- (3) **Statement-1** is incorrect but **Statement-2** is correct.
- (4) Both **Statement-1** and **Statement-2** are incorrect.

112. Statements related to human Insulin are given below. Which statement(s) is/are correct about genetically engineered Insulin?

- (a) Pro-hormone insulin contains an extra stretch of C-peptide
- (b) A-peptide and B-peptide chains of insulin were produced separately in *E.coli*, extracted and combined by creating disulphide bond between them.
- (c) Insulin used for treating diabetes was extracted from cattles and pigs.
- (d) Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone.
- (e) Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below :

- (1) (b) only
- (2) (c) and (d) only
- (3) (c), (d) and (e) only
- (4) (a), (b) and (d) only

113. The period from birth to the natural death of an organism represents its :

- (1) Life cycle
- (2) Biotic potential
- (3) Fertility
- (4) Life span

114. Fill in the blanks :

When offspring is produced by a single parent with or without the involvement of gamete formation, the reproduction is _____. When two parents participate in reproductive process and also involve fusion of male and female gametes, it is called _____ reproduction.

- (1) Sexual, asexual respectively
- (2) Asexual, sexual respectively
- (3) Vegetative, asexual respectively
- (4) Asexual, vegetative respectively

115. Select the incorrect statement ?

- (1) The number and length of stamens are variable in flowers of different species.
- (2) The distal end of filament of stamen is attached to thalamus.
- (3) A typical angiosperm anther is bilobed and dithecal.
- (4) The anther is consist of four microsporangia, two in each lobe.

116. The innermost wall layer of typical microsporangium is :

- (1) Epidermis
- (2) Middle layer
- (3) Endothecium
- (4) Tapetum

117. Cleistogamous flowers are invariably :

- (1) Geitonogamous
- (2) Xenogamous
- (3) Siphonogamous
- (4) Autogamous

118. Transfer of pollen grains from anther to the stigma of a different plant, is known as

- (1) Autogamy
- (2) Xenogamy
- (3) Geitonogamy
- (4) Self pollination

119. **Statement-I** : Complete autogamy is rather rare.

Statement-II : Autogamy requires synchrony in pollen release and stigma receptivity.

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) Both **Statement I** and **Statement II** are correct

120. Flowers secrete foul odour are pollinated by :

- (1) Wind
- (2) Water
- (3) Flies & beetles
- (4) Bees

121. Formation of fruit from unfertilized ovary is known as :

- (1) Parthenocarp
- (2) Parthenogenesis
- (3) Polyembryony
- (4) Apomixis

122. What will be the ploidy of endosperm of a seed produced after diploid female plant with tetraploid male plant ?

- (1) Tetraploid
- (2) Pentaploid
- (3) Hexaploid
- (4) Diploid

123. What is triple fusion in flowering plants ?

- (1) Fusion of male gamete with antipodals
- (2) Fusion of male gamete with secondary nucleus.
- (3) Fusion of female gamete with polar nuclei
- (4) Fusion of male gamete with female gamete

124. How many meiotic division required to form 104 male gametes ?

- (1) 104
- (2) 26
- (3) 13
- (4) 23

125. Match the column-I with column-II.

	Column-I		Column-II
A.	Egg apparatus	i.	One celled
B.	Embryo sac	ii.	Two celled
C.	Central cell	iii.	Three celled
D.	Mature pollen grains	iv.	Seven celled

Options :

- (1) A-iii, B-iv, C-i, D-ii
- (2) A-iii, B-i, C-ii, D-iv
- (3) A-iv, B-iii, C-ii, D-i
- (4) A-ii, B-i, C-iii, D-iv

126. An organic substance that can withstand environmental extremes and cannot be degraded by any enzyme is:

- (1) Lignin
- (2) Cellulose
- (3) Cuticle
- (4) Sporopollenin

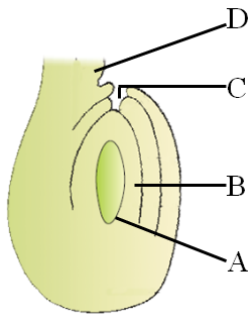
127. Inner wall of the pollen grain is made up of :

- (1) Sporopollenin
- (2) Pectin
- (3) Cellulose
- (4) Both 2 & 3

128. Which of the following sequence is correct for microsporogenesis?

- (1) Microspore tetrad (2N) $\xrightarrow{\text{Meiosis}}$ Microspore (N)
- (2) Microspore mother cell (2N) $\xrightarrow{\text{Meiosis}}$ Microspore tetrad (N)
- (3) Microspore tetrad (N) $\xrightarrow{\text{Meiosis}}$ Microspore (2N)
- (4) Microspore tetrad (2N) $\xrightarrow{\text{Meiosis}}$ Microspore (2N)

129. Find out the structure A, B, C and D in given figure :



	(A)	(B)	(C)	(D)
(1)	Embryo sac	Chalaza	Funicle	Hilum
(2)	Embryo sac	Hilum	Chalaza	Micropyle
(3)	Embryo sac	Nucellus	Funicle	Hilum
(4)	Embryo sac	Nucellus	Micropyle	Funicle

130. Given below are two statements :

Statement-I : Wind and water pollinated flowers are not very colourful and do not produce nector.

Statement-II : Majority of insect-pollinated flowers are large, colourful, fragrant and rich in nector.

Choose the correct answer from the options given below :

- (1) Both Statement-I and Statement-II are false
- (2) Statement-I is correct but Statement-II is false
- (3) Statement-I is incorrect but Statement-II is true
- (4) Both Statement-I and Statement-II are true

131. Which of the following is an example of dioecious plant?

- (1) Castor
- (2) Papaya
- (3) Maize
- (4) Cucumber

132. Which of the following statements are incorrect?

- (a) Boron and calcium elements (mainly Boron) are essential for the growth of pollen tube.
- (b) Best temperature for growth of pollen tube is 20–30 °C.
- (c) All the events from pollen deposition on the stigma until pollen tube enters the ovule are together referred to as pollen pistil interaction.
- (d) Bamboo is a polycarpic plant.

- (1) Only c
- (2) a, b, c, d
- (3) Only d
- (4) Only b, c

133. **Assertion** : Reproduction cannot be taken as defining property of livings.

Reason : Some livings can not reproduce.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false
- (3) **Assertion** is false but **Reason** is true
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

134. Sexual classification of plants is given by :

- (1) Ernst Mayr
- (2) Carolus Linnaeus
- (3) R.H. Whittaker
- (4) Robert May

135. Label on Herbarium sheet does not contain ?

- (1) Height of plant
- (2) Collector's photo
- (3) Name of Family
- (4) Both (1) & (2)

SECTION-B (BIOLOGY-I)

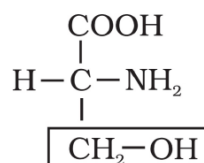
136. Which of the following statement is incorrect?

- (1) The main challenge for production of insulin using rDNA techniques was getting insulin assembled into a mature form.
- (2) If the gene isolate from marrow cells producing ADA is introduced into cells at early embryonic stages, it could be a permanent cure of ADA deficiency.
- (3) Today transgenic models exist for many human diseases such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's.
- (4) If successful and found to be reliable, mice could replace the use of donkeys to test the safety of batches of the vaccine.

137. Which two needs to be fused to form pomato?

- (1) Cells of potato and tomato
- (2) Protoplasts of potato and tomato
- (3) Mitochondria of potato and tomato
- (4) Cell walls of potato and tomato

138. Identify the following structure-



- (1) Cysteine
- (2) Glycerol
- (3) Palmitic acid
- (4) Serine

139. How many of the following varieties of wheat are developed by plant breeding techniques?

Sonalika, Pusa sawani, Kalyan sona, IR-8, Himgiri, Ratna, Pusa swarnim.

- (1) Five
- (2) Four
- (3) Three
- (4) Two

140. Which of the following statement is correct?

- (1) Chitin is not a complex polysaccharide
- (2) Cellulose gives red color in iodine test
- (3) 10-15% of total cellular mass is nucleic acids
- (4) GLUT-4 enables glucose transport into cells.

141. What term used to describe morphological and genetically similar offspring ?

- (1) Conidia
- (2) Buds
- (3) Gemmules
- (4) Clones

142. Egg apparatus consists of :

- (1) 7-cells
- (2) 2-cells
- (3) 3-cells
- (4) 4-cells

143. Amongst abiotic pollinations. Which is more common :

- (1) Wind pollination
- (2) Water pollination
- (3) Both (1) & (2)
- (4) Insect pollination

144. **Statement-I** : Majority of flowering plants produce hermaphrodite flower.

Statement-II : Among the animals, wasps are dominant biotic pollinating agents.

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) Both **Statement I** and **Statement II** are correct.

145. The nourishment to developing pollen grains is provided by :

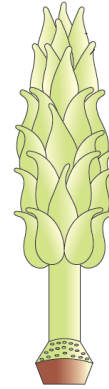
- (1) Epidermis
- (2) Endothecium
- (3) Middle layer
- (4) Tapetum

146. Arrange the following terms in the correct development sequence :

- (A) Pollen grain
- (B) Pollen mother cell
- (C) Microspore tetrad
- (D) Male gametes
- (E) Sporogenous tissue

- (1) B,D,C,E,A
- (2) E,B,C,D,A
- (3) E,B,C,A,D
- (4) E,B,A,C,D

147. Identify the given diagram and example :



- (1) Apocarpous gynoecium of *Hibiscus*.
- (2) Syncarpous gynoecium of *Michelia*.
- (3) Apocarpous gynoecium of *Michelia*
- (4) Syncarpous pistil of *Papaver*.

148. Which of the following is not a dioecious plant?

- (1) *Cucurbita*
- (2) Date palm
- (3) *Vallisneria*
- (4) Papaya

149. Wind pollinated flowers are :-

- (1) Small, producing nectar and dry pollen
- (2) Small, brightly coloured, producing large number of pollen grains
- (3) Small, producing large number of dry pollen grains
- (4) Large, producing abundant nectar and pollen

150. Transmission tissue is characteristic feature of :-

- (1) Solid style
- (2) Dry stigma
- (3) Wet stigma
- (4) Hollow style

Topic : BIOTECH-APPLICATIONS, BIOMOLECULES, PLANT BREEDING, CELL BIOLOGY (COMPLETED), PLANT DIVERSITY, COMPLETE PLANT ANATOMY [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM & REPRODUCTION IN ORGANISMS.

SECTION-A (BIOLOGY-II)

151. According to R.H. Whittaker, how many kingdom have photosynthetic organisms ?

- (1) Only one (2) Two
(3) Three (4) Four

152. Choose the correct statement for taxonomic keys :

- (A) A pair of contrasting character known as lead
(B) Each statement in keys is couplet.
(C) Keys have analytical nature
(D) Same key used for plant and animal identification

Options :

- (1) A & D only
(2) B & C only
(3) C only
(4) A, B & D only

153. 9 + 2 organisation is present in :-

- (1) Flagella of bacteria
(2) Flagella and cilia of eukaryotic cell
(3) Basal body of flagella
(4) Centriole and basal body of flagella

154. Dissolution of the synaptonemal complex occurs during :-

- (1) Leptotene (2) Pachytene
(3) Zygotene (4) Diplotene

155. Select **correct** match :

- (1) Plasma membrane - Permeable in plant cells
(2) Cell wall - Living rigid structure
(3) Primary cell wall - Capable of growth
(4) Algal cell wall - Chitin

156. Choose the incorrect statement :-

- (1) M-phase is the most dramatic period of cell cycle
(2) The metacentric chromosome has middle centromere forming two equal arms of the chromosome.
(3) The content of nucleolus is continuous with the rest of the nucleoplasm
(4) The ribosomes of the chloroplasts are smaller than the mitochondrial ribosomes.

157. The interphase nucleus :-

- (1) Nucleus of a cell when it is dividing
(2) Have chromosomes
(3) Does not have nucleolus
(4) Have pores in their nuclear envelope

158. Match the column I with column II and select the correct option from option given below :-

Column-I		Column-II	
(P)	Lysosome	(A)	Oxidation of food
(Q)	Golgi body	(B)	Protein synthesis
(R)	Mitochondria	(C)	Hydrolysis of macromolecule
(S)	Ribosome	(D)	Steroidal hormone synthesis
		(E)	Formation of glycolipid

- (1) P : E Q : D R : A S : B
(2) P : A Q : E R : C S : B
(3) P : C Q : E R : A S : B
(4) P : C Q : D R : A S : B

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159. Enzymes of Calvin cycle and protein synthesis are present in which cell organelle?

- (1) Mitochondria (2) Chloroplast
(3) Both (1) and (2) (4) Nucleus

160. The fluidity of plasma membrane increase with :

- (1) Increase in saturated fatty acids in the membrane
(2) Increase in unsaturated fatty acids in the membrane
(3) Increase in cholesterol in the membrane
(4) Increase in glycolipid content in the membrane

161. Choose the incorrect taxon for wheat :

- (1) Anacardiaceae (2) Monocotyledonae
(3) Poales (4) *Triticum*

162. Which of the following statement is incorrect for mitochondria?

- (1) Inner compartment of mitochondria termed as matrix
(2) Porin protein is present in inner membrane
(3) The cristae increase the surface area
(4) The mitochondria divide by fission

163. I division ensures the production of haploid phase in the life cycle of sexually reproducing organisms where as II restores the diploid phase.

Choose the correct word for I and II blanks :

	I	II
(1)	Mitosis	Meiosis
(2)	Mitosis	Amitosis
(3)	Amitosis	Meiosis
(4)	Meiosis	Fertilisation

164. Match the column I with column II and select the correct option given below :-

Column I		Column II	
A.	Crossing over	P.	Zygotene
B.	Chiasma	Q.	Pachytene
C.	Synaptonemal complex	R.	Diakinesis
D.	Terminalization	S.	Diplotene
		T.	Leptotene

- (1) A-Q, B-S, C-T, D-P
(2) A-S, B-Q, C-P, D-R
(3) A-Q, B-S, C-P, D-R
(4) A-S, B-Q, C-T, D-P

165. During the metaphasic mitosis :-

- (1) Chromosomal material start to condenses and form compact mitotic chromosomes
(2) Nuclear membrane, nucleolus, ER and golgi body start to disappear
(3) All the chromosomes coming to lie at the equator with one chromatid of each chromosome connected by its kinetochore to spindle fibres from one pole and its sister chromatid connected by its kinetochore to spindle fibres from opposite pole
(4) Centromere split and chromatid separate

166. Select the incorrect match :-

- (1) Anaphase - separation of chromatids
(2) Telophase - loss of identity of chromosomes
(3) Prophase - completion of condensation of chromosomes
(4) Metaphase - chromosomes have sister chromatid

167. Select the incorrect statement :-

- (1) The duration of cell cycle can vary from organism to organism and also from cell type to cell type
- (2) Yeast can progress through the cell cycle in only about 90 minutes
- (3) Of the total duration of cell cycle, the interphase lasts more than 95%
- (4) Centrioles duplicate during S phase in nucleus

168. Cells at the end of prophase when viewed under the microscope, shows :-

- (1) Golgi body
- (2) Endoplasmic reticulum
- (3) Nuclear membrane
- (4) Centrioles

169. Cell membranes possess lipid, protein and carbohydrate. The ratio of protein and lipid varies considerably in different cell types. In human beings, the membrane of the RBCs has approximately :-

- (1) 40 percent lipids and 52 percent carbohydrate
- (2) 40 percent protein and 52 percent lipids
- (3) 40 percent lipids and 52 percent proteins
- (4) 40 percent protein and 52 percent carbohydrate

170. Catalase enzyme is found in :-

- (1) ER
- (2) Golgi body
- (3) Peroxisome
- (4) Lysosome

171. At metaphase, chromosomes are attached to the spindle fibres by their :-

- (1) Centromere
- (2) Satellites
- (3) Secondary constrictions
- (4) Kinetochore

172. The size of Pleuropneumonia - like Organism (PPL0) is :

- (1) 0.02 mm
- (2) 1-2 mm
- (3) 10-20 mm
- (4) 0.1 μ m

173. Collenchyma is absent in stem of -

- (1) *Cucurbita*
- (2) Sunflower
- (3) Brinjal
- (4) Maize

174. Passage cells are thin-walled cells found in :-

- (1) Testa of seeds to enable emergence of growing embryonic axis during seed germination
- (2) Phloem elements that serve as entry points for substances for transport to other plant parts
- (3) Endodermis of roots facilitating rapid transport of water from cortex to pericycle
- (4) Central region of style through which the pollen tube grows towards the ovary.

175. Regeneration of damaged growing grass following grazing is largely due to :-

- (1) Inter fascicular cambium
- (2) Apical meristem
- (3) Intercalary meristem
- (4) Secondary meristem

176. Match **List-I** with **List-II**.

List-I		List-II	
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	pulp of sapota is made up of	(ii)	Meristematic tissue
(c)	Tissue having different types of cells	(iii)	Sclereids
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	parenchyma

Select the **correct** answer from the options given below.

	(a)	(b)	(c)	(d)
(1)	(ii)	(iv)	(i)	(iii)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(i)	(ii)	(iii)	(iv)
(4)	(iii)	(ii)	(iv)	(i)

177. Interfascicular cambium is present between :

- (1) Primary xylem and primary phloem
- (2) Pericycle and endodermis
- (3) Two vascular bundles
- (4) Secondary xylem and secondary phloem

178. In which plant vascular bundles are scattered in ground tissue and each vascular bundle is surrounded by sclerenchymatous bundle sheath?

- (1) Maize
- (2) Sunflower
- (3) Gram
- (4) *Ficus*

179. **Statement-I** : In monocot stem, the endodermis is also referred to as the starch sheath.

Statement-II : The cells of the hypodermis in root are rich in starch grains.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) **Statement-I** is incorrect but **Statement-II** is correct
- (2) Both **Statement-I** and **Statement-II** are correct
- (3) Both **Statement-I** and **Statement-II** are incorrect
- (4) **Statement-I** is correct but **Statement-II** is incorrect

180. Primary meristem is present -

- (1) At the tips of roots
- (2) At the tips of shoots
- (3) Buds present in axils of leaves
- (4) All of the above

181. Select the **correct** statement related to cork cambium in root.

- (1) The cork differentiated from cork cambium, is impervious to water.
- (2) The inner cells differentiate into phelloderm.
- (3) Cuts the cells on inner side and outer side.
- (4) All of the above.

182. The transverse section of a plant shows following anatomical features:

- (a) Large number of scattered vascular bundles surrounded by bundle sheath.
- (b) Large conspicuous parenchymatous ground tissue.
- (c) Vascular bundles conjoint and closed.
- (d) Phloem parenchyma absent.

Identify the category of plant and its part :-

- (1) Dicotyledonous root
- (2) Monocotyledonous stem
- (3) Monocotyledonous root
- (4) Dicotyledonous stem

SECTION-B (BIOLOGY-II)

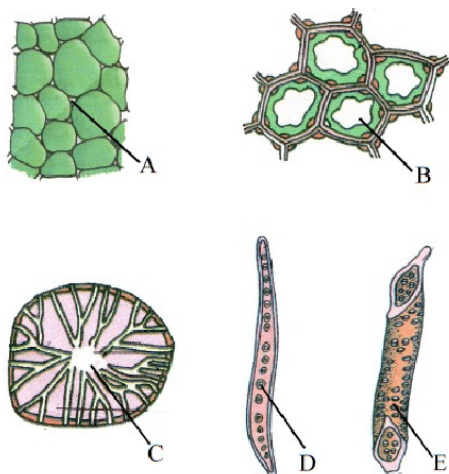
183. In monocotyledonae roots, the cells of dermatogen form a mass of small tissue, that is also meristematic and give rise to the root cap, called as :-

- (1) Plerome (2) Procambium
(3) Calyptragen (4) Periblem

184. Which one of the following is enucleated cell?

- (1) Guard cell
(2) Companion cell
(3) Vessel
(4) All the above

185. Examine the figure (A–E) given below and select the right option out of 1–4, in which all the five structure A,B,C,D & E are identified correctly.



	(A)	(B)	(C)	(D)	(E)
(1)	Inter cellular space	Vacuole	Lumen	Perforated end wall	Simple pits
(2)	Inter cellular space	Vacuole	Lumen	Bordered pits	Simple pits
(3)	Vacuole	Protoplasm	Bordered pits	Simple pits	Lumen
(4)	Perforated end wall	Thickened Corner	Pith Cavity	Lumen	Simple pits

186. **Assertion :** All livings grow.

Reason : Growth is a defining property of livings.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false
(3) **Assertion** is false but **Reason** is true
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

187. **Statement-I :** Scientific names are written in italics.

Statement-II : An organism has only one scientific name.

- (1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

188. Which of the following characters shown by all livings ?

- (A) Growth
(B) Reproduction
(C) Metabolism
(D) Cellular organization
(E) Consciousness
(F) Self consciousness

- (1) C, D, E only
(2) A, B, C, D, E and F
(3) A, C, D, E only
(4) C, D only

189. **Statement-I** : For higher plant taxonomy, sexual characters are given more importance.

Statement-II : Artificial system of classification is based on a few morphological characters.

- (1) Both **Statement I** and **Statement II** are incorrect
- (2) **Statement I** is correct but **Statement II** is incorrect
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) Both **Statement I** and **Statement II** are correct

190. **Assertion** : Higher the taxonomic category greater the difficulty to determine relationship to other taxa at same level.

Reason : At higher category common characters are less.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false
- (3) **Assertion** is false but **Reason** is true
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

191. Which of the events listed below is not observed during mitosis ?

- (1) Chromatin condensation
- (2) Movement of centrioles to opposite poles
- (3) Appearance of chromosomes with two chromatids joined together at the centromere
- (4) Crossing over

192. Endoplasmic reticulum do not involved in :-

- (1) Transportation of molecules
- (2) Glycogen synthesis
- (3) Fat synthesis
- (4) Pectin synthesis

193. If the pollen grain of plant has 20 pg of DNA in its nucleus then how much DNA would a leaf cell of same organism have in metaphase of mitosis ?

- (1) 40 pg
- (2) 20 pg
- (3) 160 pg
- (4) 80 pg

194. Select the incorrect statement :-

- (1) Cell plate formation occurs in plant cell during cytokinesis
- (2) Direction of cytokinesis in plant cell is centripetal
- (3) Liquid endosperm of coconut is formed by free nuclear division
- (4) Peripheral contractile ring formation occurs in animal cell during cytokinesis

195. During mitosis ER and nucleolus begin to disappear at :

- (1) Early prophase
- (2) Late prophase
- (3) Early metaphase
- (4) Late metaphase

196. Sclerenchyma differ from collenchyma in -

- (1) Devoid protoplast at maturity
- (2) Having thin walls
- (3) Having lignin
- (4) both (1) and (3)

197. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R** :

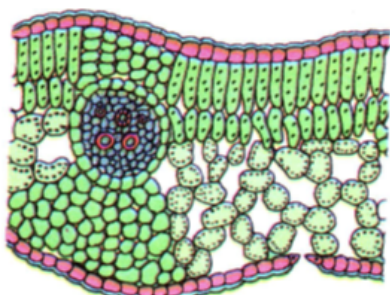
Assertion A : Late wood has fewer xylary elements with narrow vessels.

Reason R : Vascular cambium is less active in winters.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

198. Observe the given diagram carefully and how many statements are true about given diagram :



- (a) The position of xylem in the vascular bundle is adaxial
- (b) The position of phloem in the vascular bundle is abaxial
- (c) Two distinct patches of parenchyma are present above & below of the large vascular bundle.
- (d) The adaxially placed palisade parenchyma is made up of elongated cells which are arranged vertically and parallel to each other.

- (1) Two
- (2) Three
- (3) Four
- (4) One

199. In sunflower stem, the primary xylem is called endarch. Because ?

- (1) the protoxylem is surrounded by the metaxylem
- (2) both the protoxylem and the metaxylem lie towards the centre
- (3) the protoxylem lies towards the centre and the metaxylem lies towards the periphery.
- (4) the protoxylem lies towards the periphery and the metaxylem lies towards the centre

200. Match the column-I with column-II and choose correct option.

Column-I		Column-II	
(a)	Xylem fibres	(i)	Chisel like ends and pitted end wall
(b)	Xylem vessels	(ii)	Store food material
(c)	Xylem tracheids	(iii)	Obliterated central lumen
(d)	Xylem parenchyma	(iv)	Perforated end wall

- (1) (a)-i, (b)-ii, (c)-iii, (d)-iv
- (2) (a)-iv, (b)-iii, (c)-i, (d)-ii
- (3) (a)-iii, (b)-iv, (c)-i, (d)-ii
- (4) (a)-iii, (b)-i, (c)-iv, (d)-ii

 QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)			
ENTHUSE ADV. COURSE : MEF			
MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
07 10-09-2023	SYLLABUS COVERED FROM 16-08-2023 TO 09-09-2023	P	WAVE OPTICS AND U & D, ERROR
		C	GOC - I
		B	ECOLOGY

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